

Allocating Causal Contribution to Market Externalities: Shapley Attribution, Elasticity Pivotality, and Pigouvian Counterfactual Analysis

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Abstract—When a competitive market outcome generates a harmful externality (e.g., CO₂ emissions), it is unclear how to distribute causal contribution to realized harm among heterogeneous producers and consumers. This paper proposes three complementary, curve-based frameworks. First, a cooperative-game attribution assigns each actor a share of realized harm via the Shapley value of a characteristic function induced by coalition-specific market clearing. Second, a local (marginal) pivotality framework allocates realized harm using comparative statics: each agent’s contribution is proportional to the sensitivity of equilibrium quantity to a small, proportional scaling of that agent’s supply or demand curve; this yields an explicit elasticity-based decomposition. Third, a Pigouvian-counterfactual framework distinguishes (i) unavoidable harm at the welfare-optimal (internalized) outcome from (ii) avoidable excess harm due to underpricing the externality, and allocates the latter according to private gains from the uncorrected equilibrium. The paper develops all three approaches for generic (nonlinear) supply and demand functions, and illustrates symbolic worked examples for two producers and three consumers. We also discuss how imperfect information about others’ curves motivates *ex ante* (Bayesian) contribution measures, and we emphasize throughout that all three quantities are measures of causal contribution to an outcome, not measures of moral blame or culpability.

I. INTRODUCTION

Markets coordinate production and consumption through prices, but when traded goods generate harmful externalities, the equilibrium quantity can exceed the socially desirable level. Beyond standard policy prescriptions (taxes, caps, regulation), there is a normative question: *how should causal contribution to realized harm be distributed between producers and consumers*, especially when each agent has a different supply or demand curve?

This paper formalizes three complementary answers based on primitives familiar from microeconomic theory. The first is an *attribution* approach: causal contribution is allocated for the realized harm produced by the actual market outcome, using a cooperative-game interpretation and Shapley values. The second is a *local pivotality* approach: causal contribution is allocated for realized harm based on each agent’s marginal influence on equilibrium quantity under a small proportional perturbation of their curve, yielding an elasticity-based split between the producer and consumer sides. The third is a *counterfactual* approach: causal contribution to *avoidable* harm is allocated by comparing the unregulated equilibrium to a

Pigouvian benchmark and distributing the resulting harm gap according to the private benefits of underpricing.

A key theme is that a causal-contribution measure depends on (i) a harm model, (ii) an equilibrium/market-clearing institution, and (iii) a counterfactual baseline. We keep the environment simple (single good, competitive behavior, quantity-only harm) to highlight what is structural versus normative.

A. Causal Contribution versus Moral Blame

Before proceeding, it is worth being precise about what the quantities defined in this paper do and do not represent. Each of the three frameworks below takes a fixed, realized (or counterfactually specified) harm outcome and decomposes it into agent-level shares that sum back to the total. Formally, this is an exercise in *causal contribution*: given a model of how the market outcome depends on each agent’s participation, how much of that outcome traces back to each agent, under some well-defined counterfactual or marginal comparison (coalition averaging for Shapley, local comparative statics for pivotality, benefit from a policy gap for the Pigouvian measure). None of these constructions makes reference to an agent’s knowledge, intentions, available alternatives, or the reasonableness of their conduct.

Moral blame (or culpability, fault, responsibility in the everyday normative sense) is a different and stricter notion. Two agents who make an identical causal contribution to harm are not automatically equally blameworthy: blame typically also depends on whether the agent could foreseeably have acted otherwise, whether a less harmful alternative was reasonably available to them, whether they possessed relevant information, and whether their conduct fell short of some standard of care. A producer who causes a given quantity of emissions using the best available technology and one who causes the identical quantity using needlessly dirty technology despite having an affordable cleaner alternative make the same causal contribution to the harm outcome, in the sense measured here, but few moral frameworks would treat them as equally at fault. Conversely, an agent with a very large causal contribution (e.g., a highly inelastic consumer who is nearly powerless to reduce their participation) may attract little blame precisely because they lacked a realistic alternative.

Because all three constructions in this paper are purely structural decompositions of an outcome, we use *causal contribution* throughout rather than *blame* or *culpability*, and we

reserve the latter terms for the additional, separate judgment – concerning fault, avoidability, and agency – that would be required to turn a causal-contribution share into a genuine attribution of moral responsibility. Section VI discusses one natural way to begin bridging this gap by conditioning contribution measures on an agent's available information.

II. SETUP

A. Agents and curves

There are N producers indexed by $i \in \mathcal{P} = \{1, \dots, N\}$ and M consumers indexed by $j \in \mathcal{C} = \{1, \dots, M\}$. Let the set of all agents be $\mathcal{N} = \mathcal{P} \cup \mathcal{C}$.

a) *Supply.*: Producer i supplies

$$s_i(p) \geq 0, \quad s'_i(p) \geq 0, \quad (1)$$

for price $p \geq 0$.

b) *Demand.*: Consumer j demands

$$d_j(p) \geq 0, \quad d'_j(p) \leq 0. \quad (2)$$

c) *Aggregation.*: Aggregate supply and demand are

$$S(p) = \sum_{i \in \mathcal{P}} s_i(p), \quad D(p) = \sum_{j \in \mathcal{C}} d_j(p). \quad (3)$$

B. External harm

We assume a *quantity-only* externality: total external harm depends on total traded quantity Q via

$$H(Q), \quad H(0) = 0, \quad H'(Q) \geq 0. \quad (4)$$

We interpret $H(Q)$ as monetized damages, or any agreed scalar measure of harm. The marginal harm is

$$MH(Q) \triangleq H'(Q). \quad (5)$$

C. Competitive equilibrium

A (Walrasian) competitive equilibrium price p^* solves

$$S(p^*) = D(p^*), \quad (6)$$

with equilibrium quantity

$$Q^* = S(p^*) = D(p^*), \quad (7)$$

and realized harm

$$H^* \triangleq H(Q^*). \quad (8)$$

III. SHAPLEY ATTRIBUTION FOR REALIZED HARM

A. Characteristic function via coalition markets

To allocate causal contribution to realized harm, we define a cooperative game where the *value* of a coalition is the harm it generates when only members of the coalition participate in the market.

Let $T \subseteq \mathcal{N}$ be a coalition. Define coalition supply and demand:

$$S_T(p) = \sum_{i \in T \cap \mathcal{P}} s_i(p), \quad D_T(p) = \sum_{j \in T \cap \mathcal{C}} d_j(p). \quad (9)$$

A coalition market-clearing price $p(T)$ solves

$$S_T(p(T)) = D_T(p(T)), \quad (10)$$

whenever T contains at least one producer and one consumer and a clearing price exists. Define the coalition quantity as

$$Q(T) = S_T(p(T)) = D_T(p(T)). \quad (11)$$

For coalitions that cannot trade (no producers or no consumers, or no clearing), we set $Q(T) = 0$.

a) *Quantity-only harm game.*: Define the characteristic function

$$v(T) \triangleq H(Q(T)). \quad (12)$$

This encodes the idea that harm is jointly produced by the coalition's capacity to supply and willingness to demand.

B. Shapley value

The Shapley value assigns each agent $a \in \mathcal{N}$ a causal-contribution share

$$\phi_a(v) = \sum_{T \subseteq \mathcal{N} \setminus \{a\}} \frac{|T|! (|\mathcal{N}| - |T| - 1)!}{|\mathcal{N}|!} (v(T \cup \{a\}) - v(T)). \quad (13)$$

Interpretation: $\phi_a(v)$ is the expected marginal increase in harm caused by adding agent a to a random ordering of all agents.

a) *Budget balance.*: Because $v(\emptyset) = 0$, the Shapley value satisfies

$$\sum_{a \in \mathcal{N}} \phi_a(v) = v(\mathcal{N}) = H(Q^*) = H^*, \quad (14)$$

so total causal contribution equals realized harm.

C. Quantity-only specialization

A commonly used special case sets

$$H(Q) = \kappa Q, \quad \kappa > 0, \quad (15)$$

so that $v(T) = \kappa Q(T)$. Then Shapley contribution is linear in coalition quantities:

$$\phi_a(v) = \kappa \cdot \phi_a(Q), \quad (16)$$

where $\phi_a(Q)$ is computed from (13) with $v(T) = Q(T)$.

D. Closed-form Shapley attribution for identical linear agents

This subsection provides an explicit Shapley attribution in a tractable special case: N_P identical producers and N_C identical consumers with linear (truncated) curves, and quantity-only harm. The goal is to make the dependence of the producer-consumer split on slope parameters transparent.

a) *Identical truncated-linear curves.*: Let each producer have supply

$$q^s(p) = \max\{0, \alpha p - \gamma\}, \quad \alpha > 0, \quad (17)$$

and each consumer have demand

$$q^d(p) = \max\{0, \delta - \beta p\}, \quad \beta > 0. \quad (18)$$

Define the overlap parameter

$$X \triangleq \alpha\delta - \beta\gamma. \quad (19)$$

If $X \leq 0$ (equivalently $\delta/\beta \leq \gamma/\alpha$), there is no price at which both sides are simultaneously positive, so every coalition trades 0 and all Shapley contributions are 0. Below we assume $X > 0$.

b) *Coalition equilibrium quantity.*: For any coalition containing $m \in \{0, 1, \dots, N_P\}$ producers and $n \in \{0, 1, \dots, N_C\}$ consumers, aggregate (untruncated) supply and demand are

$$Q_m^s(p) = m(\alpha p - \gamma), \quad Q_n^d(p) = n(\delta - \beta p). \quad (20)$$

For $m, n > 0$, coalition market clearing yields

$$p(m, n) = \frac{n\delta + m\gamma}{m\alpha + n\beta}, \quad (21)$$

and the cleared coalition quantity is

$$Q(m, n) = \frac{mn(\alpha\delta - \beta\gamma)}{m\alpha + n\beta} = \frac{mnX}{m\alpha + n\beta}. \quad (22)$$

When $X > 0$, the clearing price $p(m, n)$ lies between γ/α and δ/β , so truncation does not bind and (22) is valid. We therefore define the quantity-only characteristic function

$$v(m, n) = \begin{cases} 0, & m = 0 \text{ or } n = 0, \\ H(Q(m, n)), & m, n > 0, \end{cases} \quad (23)$$

and in the linear-harm case (15) this reduces to $v(m, n) = \kappa Q(m, n)$.

c) *Marginal contribution of an additional producer.*: Fix a particular producer. Consider a predecessor set in a Shapley permutation containing

$$\begin{aligned} i &\in \{0, \dots, N_P - 1\} \text{ other producers,} \\ j &\in \{0, \dots, N_C\} \text{ consumers.} \end{aligned}$$

For $j = 0$, trade is impossible and the marginal contribution is zero. For $j \geq 1$, the marginal increase in coalition quantity from adding one producer is

$$\begin{aligned} \Delta_P(i, j) &\triangleq Q(i + 1, j) - Q(i, j) \\ &= \frac{j^2 \beta X}{(i\alpha + j\beta)((i + 1)\alpha + j\beta)}. \end{aligned} \quad (24)$$

d) *Exact Shapley value for one producer (finite sum).*: Let $n_{\text{tot}} = N_P + N_C$ be the total number of agents. In the Shapley formula (13), each predecessor set T with $|T| = k$ carries weight

$$\frac{k! (n_{\text{tot}} - k - 1)!}{n_{\text{tot}}!} = \frac{1}{n_{\text{tot}}} \cdot \frac{1}{\binom{n_{\text{tot}} - 1}{k}}. \quad (25)$$

The number of predecessor sets with (i, j) composition is $\binom{N_P - 1}{i} \binom{N_C}{j}$, and $k = i + j$. Thus, for linear harm $H(Q) = \kappa Q$ the Shapley causal contribution of one producer is

$$\phi_P = \kappa \sum_{i=0}^{N_P - 1} \sum_{j=1}^{N_C} \binom{N_P - 1}{i} \binom{N_C}{j} \frac{1}{n_{\text{tot}}} \frac{1}{\binom{n_{\text{tot}} - 1}{i + j}} \Delta_P(i, j), \quad (26)$$

with $\Delta_P(i, j)$ given by (24). By symmetry, all producers receive the same ϕ_P .

e) *Total producer-side Shapley contribution.*: The combined Shapley causal contribution of all producers is

$$B_P^{\text{Sh}} = N_P \phi_P, \quad (27)$$

and budget balance implies $B_P^{\text{Sh}} + B_C^{\text{Sh}} = \kappa Q(N_P, N_C)$ where

$$Q(N_P, N_C) = \frac{N_P N_C X}{N_P \alpha + N_C \beta}. \quad (28)$$

It is often convenient to consider the producer share of realized-harm causal contribution,

$$s_P^{\text{Sh}} \triangleq \frac{B_P^{\text{Sh}}}{\kappa Q(N_P, N_C)}. \quad (29)$$

Because both numerator and denominator are proportional to X , the share s_P^{Sh} depends on γ, δ only through the feasibility condition $X > 0$, and otherwise depends on (N_P, N_C) and the slope ratio

$$r \triangleq \frac{\alpha}{\beta}. \quad (30)$$

f) *Simple limiting cases.*: The Shapley share (29) admits transparent limits as r varies.

- **Inelastic supply** ($r \rightarrow 0$). In this limit, for $j \geq 1$ the coalition quantity becomes essentially independent of j :

$$Q(m, n) = \frac{mnX}{m\alpha + n\beta} \xrightarrow{r \rightarrow 0} \frac{mX}{\beta} \quad (n > 0), \quad (31)$$

so each additional producer adds approximately X/β once any consumer is present. Under random arrival orderings, the expected fraction of producers arriving after the first consumer is $N_C/(N_C + 1)$, yielding

$$s_P^{\text{Sh}} \xrightarrow{r \rightarrow 0} \frac{N_C}{N_C + 1}, \quad 1 - s_P^{\text{Sh}} \xrightarrow{r \rightarrow 0} \frac{1}{N_C + 1}. \quad (32)$$

- **Elastic supply** ($r \rightarrow \infty$). In this limit, for $m \geq 1$ the coalition quantity becomes essentially independent of m :

$$Q(m, n) \xrightarrow{r \rightarrow \infty} \frac{nX}{\alpha} \quad (m > 0), \quad (33)$$

so only the first producer to arrive matters. The producer-side Shapley share tends to

$$s_P^{\text{Sh}} \xrightarrow{r \rightarrow \infty} \frac{1}{N_P + 1}, \quad 1 - s_P^{\text{Sh}} \xrightarrow{r \rightarrow \infty} \frac{N_P}{N_P + 1}. \quad (34)$$

- **Symmetry point** ($N_P = N_C$ and $r = 1$). When the two groups have equal counts and equal slopes ($\alpha = \beta$), the value function $Q(m, n)$ is invariant under exchanging the labels “producer” and “consumer”. By Shapley symmetry, the total causal contribution must split evenly:

$$N_P = N_C, r = 1 \Rightarrow s_P^{\text{Sh}} = \frac{1}{2}. \quad (35)$$

g) *Remark (comparison to elasticity pivotality).*: The limiting formulas (32)–(34) provide a global (order-averaged) notion of producer pivotality. By contrast, Section IV shows that local pivotality at the observed equilibrium yields producer-side share $\Sigma_D/(\Sigma_S + \Sigma_D)$; in linear symmetric environments these two measures often agree qualitatively, but generally differ because Shapley attribution averages marginal contributions across the entire coalition lattice rather than taking a local derivative at the grand-coalition equilibrium.

E. Remarks

The Shapley attribution depends on (i) the coalition trading rule (here, coalition-specific competitive equilibrium) and (ii) the harm function. It allocates realized harm without invoking a policy baseline. As emphasized in Section I-A, it corresponds to a cooperative *causal contribution* concept: who is pivotal for the market outcome that generates harm, averaged across coalition orderings – not a judgment about fault or culpability. Section IV develops a complementary local (marginal) pivotality measure that yields closed-form expressions in terms of slopes and elasticities at the observed equilibrium.

IV. LOCAL PIVOTALITY ATTRIBUTION VIA ELASTICITIES

Shapley values allocate realized harm by averaging marginal contributions across all coalition orderings. A complementary approach keeps the full market fixed and measures *local pivotality*: how sensitive the equilibrium quantity is to a small, proportional change in a given agent’s participation. This produces a causal-contribution measure that is (i) curve-based, (ii) interpretable via comparative statics and incidence, and (iii) computable from local slope/elasticity information at equilibrium.

A. Proportional participation perturbations

Introduce participation (scaling) parameters $\lambda_i > 0$ for producers and $\mu_j > 0$ for consumers:

$$S(p; \lambda) = \sum_{i \in \mathcal{P}} \lambda_i s_i(p), \quad D(p; \mu) = \sum_{j \in \mathcal{C}} \mu_j d_j(p), \quad (36)$$

with baseline $\lambda_i = \mu_j = 1$ for all i, j . The (perturbed) market-clearing condition is

$$F(p; \lambda, \mu) \triangleq S(p; \lambda) - D(p; \mu) = 0. \quad (37)$$

Define log-parameters $\theta_i \triangleq \ln \lambda_i$ and $\psi_j \triangleq \ln \mu_j$. A differential $d\theta_i$ is an equal proportional (percent) perturbation across agents.

Let $p(\lambda, \mu)$ be the clearing price solving (37), and define the corresponding cleared quantity

$$Q(\lambda, \mu) \triangleq S(p(\lambda, \mu); \lambda) = D(p(\lambda, \mu); \mu). \quad (38)$$

We write $p^* = p(1, 1)$ and $Q^* = Q(1, 1)$ for the baseline equilibrium.

B. Comparative statics at equilibrium

Assume differentiability and that the implicit-function regularity condition holds at $(p^*, 1, 1)$:

$$F_p(p^*; 1, 1) = S'(p^*) - D'(p^*) > 0, \quad (39)$$

where $S'(p) = \sum_i s'_i(p)$ and $D'(p) = \sum_j d'_j(p) \leq 0$. Define the positive slope aggregates

$$A \triangleq S'(p^*) = \sum_{i \in \mathcal{P}} s'_i(p^*), \quad B \triangleq -D'(p^*) = -\sum_{j \in \mathcal{C}} d'_j(p^*), \quad (40)$$

so that $F_p = A + B$.

Let each producer’s equilibrium quantity be $q_i^* \triangleq s_i(p^*)$ and each consumer’s equilibrium quantity be $x_j^* \triangleq d_j(p^*)$, so $\sum_i q_i^* = \sum_j x_j^* = Q^*$.

a) *Producer pivotality.*: Differentiate (37) with respect to θ_i at baseline. Since $\partial S/\partial \theta_i = \lambda_i s_i(p)$ and $\partial D/\partial \theta_i = 0$,

$$\left. \frac{\partial p}{\partial \theta_i} \right|_* = -\frac{q_i^*}{A + B}. \quad (41)$$

The derivative of cleared quantity with respect to θ_i is

$$\left. \frac{\partial Q}{\partial \theta_i} \right|_* = \underbrace{q_i^*}_{\text{direct scale}} + \underbrace{A \left. \frac{\partial p}{\partial \theta_i} \right|_*}_{\text{price feedback}} = q_i^* \left(1 - \frac{A}{A + B} \right) = q_i^* \frac{B}{A + B}. \quad (42)$$

b) *Consumer pivotality.*: Similarly, for ψ_j we have $\partial D/\partial \psi_j = \mu_j d_j(p)$ and $\partial S/\partial \psi_j = 0$, so

$$\left. \frac{\partial p}{\partial \psi_j} \right|_* = +\frac{x_j^*}{A + B}, \quad (43)$$

and the cleared quantity response is

$$\left. \frac{\partial Q}{\partial \psi_j} \right|_* = A \left. \frac{\partial p}{\partial \psi_j} \right|_* = x_j^* \frac{A}{A + B}. \quad (44)$$

c) *Normalization.*: Summing (42) and (44) over all agents yields

$$\sum_{i \in \mathcal{P}} \left. \frac{\partial Q}{\partial \theta_i} \right|_* + \sum_{j \in \mathcal{C}} \left. \frac{\partial Q}{\partial \psi_j} \right|_* = \frac{B}{A + B} \sum_i q_i^* + \frac{A}{A + B} \sum_j x_j^* = Q^*, \quad (45)$$

so pivotality weights aggregate to total quantity.

C. Elasticity representation

Define individual point elasticities at equilibrium:

$$\eta_i \triangleq \frac{p^*}{q_i^*} s'_i(p^*) \quad (> 0), \quad \varepsilon_j \triangleq -\frac{p^*}{x_j^*} d'_j(p^*) \quad (> 0). \quad (46)$$

Then

$$A = \frac{1}{p^*} \sum_{i \in \mathcal{P}} \eta_i q_i^*, \quad B = \frac{1}{p^*} \sum_{j \in \mathcal{C}} \varepsilon_j x_j^*. \quad (47)$$

Define

$$\Sigma_S \triangleq \sum_{i \in \mathcal{P}} \eta_i q_i^*, \quad \Sigma_D \triangleq \sum_{j \in \mathcal{C}} \varepsilon_j x_j^*. \quad (48)$$

Then $\frac{B}{A+B} = \frac{\Sigma_D}{\Sigma_S + \Sigma_D}$ and $\frac{A}{A+B} = \frac{\Sigma_S}{\Sigma_S + \Sigma_D}$.

D. Pivotality-based causal-contribution allocation for realized harm

Let pivotality weights be the quantity sensitivities:

$$w_{P_i} \triangleq \left. \frac{\partial Q}{\partial \theta_i} \right|_*, \quad w_{C_j} \triangleq \left. \frac{\partial Q}{\partial \psi_j} \right|_*. \quad (49)$$

A natural realized-harm allocation assigns shares proportional to these weights:

$$\beta_a \triangleq H^* \cdot \frac{w_a}{Q^*}, \quad a \in \mathcal{N}. \quad (50)$$

By (45), the allocation is budget balanced: $\sum_{a \in \mathcal{N}} \beta_a = H^*$.

Combining (42)–(44) with (50), we obtain explicit formulas:

$$\beta_{P_i} = H^* \cdot \frac{q_i^*}{Q^*} \cdot \frac{B}{A+B} = H^* \cdot \frac{q_i^*}{Q^*} \cdot \frac{\Sigma_D}{\Sigma_S + \Sigma_D}, \quad i \in \mathcal{P}, \quad (51)$$

$$\beta_{C_j} = H^* \cdot \frac{x_j^*}{Q^*} \cdot \frac{A}{A+B} = H^* \cdot \frac{x_j^*}{Q^*} \cdot \frac{\Sigma_S}{\Sigma_S + \Sigma_D}, \quad j \in \mathcal{C}. \quad (52)$$

Thus, the producer-side share of realized-harm causal contribution is $\Sigma_D/(\Sigma_S + \Sigma_D)$ and the consumer-side share is $\Sigma_S/(\Sigma_S + \Sigma_D)$, with within-side allocations proportional to equilibrium quantities.

V. PIGOUVIAN COUNTERFACTUAL CONTRIBUTION TO AVOIDABLE HARM

Shapley attribution and elasticity pivotality allocate the harm produced by the actual outcome. A different question is: *how much of that harm is avoidable given the existence of a welfare-improving internalization policy, and who contributes causally to the avoidable portion?* Pigouvian analysis provides a canonical counterfactual.

A. Welfare and the efficient quantity

Assume aggregate demand and supply are invertible (at least locally) so we can define inverse demand and supply:

$$p_D(Q) = D^{-1}(Q), \quad p_S(Q) = S^{-1}(Q), \quad (53)$$

with $p'_D(Q) \leq 0$, $p'_S(Q) \geq 0$.

Define utilitarian social welfare (quasi-linear, no income effects) as

$$W(Q) = \int_0^Q p_D(q) dq - \int_0^Q p_S(q) dq - H(Q). \quad (54)$$

An interior welfare optimum Q^{soc} satisfies the first-order condition

$$p_D(Q^{\text{soc}}) = p_S(Q^{\text{soc}}) + MH(Q^{\text{soc}}). \quad (55)$$

This is the standard *marginal benefit equals marginal social cost* rule.

B. Pigouvian tax schedule and implementation

Consider a per-unit tax (or fee) t that creates a wedge between consumer and producer prices:

$$p_c - p_p = t. \quad (56)$$

With a constant tax t , equilibrium must satisfy

$$S(p_p) = D(p_c), \quad p_c = p_p + t. \quad (57)$$

Equivalently, in p_c alone:

$$S(p_c - t) = D(p_c). \quad (58)$$

In general, if marginal harm varies with Q , the Pigouvian instrument is a *schedule*:

$$t(Q) = MH(Q). \quad (59)$$

At any implemented quantity Q , market equilibrium implies $p_c = p_D(Q)$ and $p_p = p_S(Q)$, and thus

$$p_D(Q) - p_S(Q) = t(Q). \quad (60)$$

Setting $t(Q) = MH(Q)$ yields (55), implementing Q^{soc} under standard regularity assumptions.

C. Avoidable harm and the counterfactual gap

Define the Pigouvian benchmark harm

$$H^{\text{soc}} \triangleq H(Q^{\text{soc}}), \quad (61)$$

and the *avoidable (excess) harm* due to underpricing as

$$\Delta H \triangleq H(Q^*) - H(Q^{\text{soc}}) = H^* - H^{\text{soc}} \geq 0. \quad (62)$$

The Pigouvian counterfactual does *not* declare H^{soc} morally “fine” in an absolute sense; rather, it is the harm level consistent with maximizing (54) under the model’s welfare criterion. The counterfactual gap ΔH isolates harm that a marginal internalization policy would remove.

D. Pigouvian causal contribution via private gains from underpricing

One principled contribution rule is to allocate ΔH in proportion to each agent’s private surplus *gain* from moving from the Pigouvian outcome to the unregulated outcome.

Let each consumer j have (quasi-linear) indirect surplus at price p :

$$CS_j(p) = \int_p^\infty d_j(x) dx, \quad (63)$$

assuming the integral exists (e.g., compact support or sufficiently fast decay). Let each producer i have producer surplus at producer price p :

$$PS_i(p) = \int_0^p s_i(x) dx - \int_0^{s_i(p)} MC_i(q) dq, \quad (64)$$

or, equivalently, when supply is derived from a convex cost function $C_i(q)$:

$$PS_i(p) = p s_i(p) - C_i(s_i(p)). \quad (65)$$

Let $(p_c^{\text{soc}}, p_p^{\text{soc}})$ denote the consumer and producer prices at the Pigouvian-implemented allocation (with wedge $t(Q^{\text{soc}})$),

and let p^* denote the unregulated market price (where $p_c = p_p = p^*$).

Define each agent's private gain from underpricing:

$$\Delta CS_j \triangleq CS_j(p^*) - CS_j(p_c^{\text{soc}}), \quad (66)$$

$$\Delta PS_i \triangleq PS_i(p^*) - PS_i(p_p^{\text{soc}}). \quad (67)$$

Let total private gain be

$$\Delta \Pi \triangleq \sum_{j \in \mathcal{C}} \Delta CS_j + \sum_{i \in \mathcal{P}} \Delta PS_i. \quad (68)$$

a) *Pigouvian causal-contribution allocation*.: Define contribution shares

$$B_a \triangleq \Delta H \cdot \frac{\Delta \Pi_a}{\Delta \Pi}, \quad \Delta \Pi_a = \begin{cases} \Delta PS_a, & a \in \mathcal{P}, \\ \Delta CS_a, & a \in \mathcal{C}. \end{cases} \quad (69)$$

Then

$$\sum_{a \in \mathcal{N}} B_a = \Delta H. \quad (70)$$

E. Interpretation and variants

Equation (69) ties avoidable-harm causal contribution to who privately benefits from the uninternalized externality; note that this is best read as an *unjust-enrichment*-style contribution rule rather than a fault-based one, since it says nothing about whether an agent could have known or acted otherwise. Alternative Pigouvian contribution notions include allocating by tax incidence (who bears welfare losses under the Pigouvian policy), by marginal contributions to the wedge, or by ability-to-abate. The key structural step is always the decomposition

$$H^* = H^{\text{soc}} + \Delta H, \quad (71)$$

and deciding which component (or both) is the object of allocation.

VI. IMPERFECT INFORMATION: EX ANTE CONTRIBUTION

So far, curves are treated as known objects for the analyst. In many settings, agents do not know others' supply/demand schedules; they may observe only prices. This motivates an *ex ante* perspective, which also offers one natural route toward the fault-sensitive notion of responsibility discussed in Section I-A: conditioning contribution on what an agent could plausibly have known moves the measure a step closer to culpability, though it still stops short of a full account of blame, which would also need to consider available alternatives and standards of care.

Let each producer have a private type θ_i determining $s_i(p; \theta_i)$ and each consumer a private type η_j determining $d_j(p; \eta_j)$. Let $\omega = (\theta_1, \dots, \theta_N, \eta_1, \dots, \eta_M)$ be the type profile drawn from a common prior $\mathbb{P}$, and let $v_\omega(\cdot)$ be the realized characteristic function induced by ω .

a) *Bayesian realized-harm attribution*.: For any realized-harm causal-contribution rule $R_a(\omega)$ (e.g., Shapley $\phi_a(v_\omega)$, or pivotality $\beta_a(\omega)$ computed from (51)–(52)), define ex ante contribution as the expectation:

$$\Phi_a \triangleq \mathbb{E}_\omega[R_a(\omega)]. \quad (72)$$

b) *Information-conditioned culpability*.: If agent a has information $\mathcal{I}_a$ (e.g., own type and observed price), a culpability-oriented measure can be defined by conditioning:

$$\Phi_a^{\text{info}} \triangleq \mathbb{E}[R_a(\omega) \mid \mathcal{I}_a], \quad (73)$$

possibly supplemented by a risk-sensitive term capturing foreseeability of large contributions. This separates *causal contribution* (ex post) from *foreseeable responsibility* (ex ante); neither, on its own, is yet a complete account of moral blame in the sense of Section I-A.

VII. SYMBOLIC WORKED EXAMPLES (TWO PRODUCERS, THREE CONSUMERS)

This section illustrates the mechanics with symbols only, for the case of two producers and three consumers:

$$\mathcal{P} = \{P_1, P_2\}, \quad (74)$$

$$\mathcal{C} = \{C_1, C_2, C_3\}, \quad (75)$$

$$\mathcal{N} = \{P_1, P_2, C_1, C_2, C_3\}. \quad (76)$$

A. Generic coalition quantities

For any coalition $T \subseteq \mathcal{N}$, define

$$S_T(p) = \sum_{P_i \in T} s_i(p), \quad D_T(p) = \sum_{C_j \in T} d_j(p). \quad (77)$$

Coalition market clearing yields $p(T)$ such that $S_T(p(T)) = D_T(p(T))$ and

$$Q(T) = S_T(p(T)) = D_T(p(T)). \quad (78)$$

The Shapley game is $v(T) = H(Q(T))$.

B. Shapley contribution: explicit expansion for P_1

With $|\mathcal{N}| = 5$, the Shapley value for P_1 is

$$\phi_{P_1} = \sum_{T \subseteq \mathcal{N} \setminus \{P_1\}} \frac{|T|!(4-|T|)!}{5!} (H(Q(T \cup \{P_1\})) - H(Q(T))). \quad (79)$$

Writing $\mathcal{N} \setminus \{P_1\} = \{P_2, C_1, C_2, C_3\}$, we can group terms by $k = |T|$:

$$\phi_{P_1} = \sum_{k=0}^4 w_k \sum_{\substack{|T|=k \\ T \subseteq \{P_2, C_1, C_2, C_3\}}} (H(Q(T \cup \{P_1\})) - H(Q(T))), \quad (80)$$

$$w_k \triangleq \frac{k!(4-k)!}{5!}. \quad (81)$$

This expression is fully symbolic; evaluation reduces to computing the relevant coalition equilibrium quantities $Q(T)$.

C. Elasticity pivotality: explicit realized-harm shares

Let (p^*, Q^*) be the unregulated equilibrium. Define equilibrium individual quantities

$$q_1^* = s_1(p^*), \quad q_2^* = s_2(p^*), \quad (82)$$

$$x_1^* = d_1(p^*), \quad x_2^* = d_2(p^*), \quad x_3^* = d_3(p^*), \quad (83)$$

with $Q^* = q_1^* + q_2^* = x_1^* + x_2^* + x_3^*$. Define point elasticities

$$\eta_1 = \frac{p^*}{q_1^*} s_1'(p^*), \quad \eta_2 = \frac{p^*}{q_2^*} s_2'(p^*), \quad (84)$$

$$\varepsilon_j = -\frac{p^*}{x_j^*} d_j'(p^*), \quad j = 1, 2, 3. \quad (85)$$

Let $\Sigma_S = \eta_1 q_1^* + \eta_2 q_2^*$ and $\Sigma_D = \varepsilon_1 x_1^* + \varepsilon_2 x_2^* + \varepsilon_3 x_3^*$. Then the pivotality-based realized-harm allocation (51)–(52) gives

$$\beta_{P_1} = H^* \cdot \frac{q_1^*}{Q^*} \cdot \frac{\Sigma_D}{\Sigma_S + \Sigma_D}, \quad (86)$$

$$\beta_{P_2} = H^* \cdot \frac{q_2^*}{Q^*} \cdot \frac{\Sigma_D}{\Sigma_S + \Sigma_D}, \quad (87)$$

$$\beta_{C_1} = H^* \cdot \frac{x_1^*}{Q^*} \cdot \frac{\Sigma_S}{\Sigma_S + \Sigma_D}, \quad (88)$$

$$\beta_{C_2} = H^* \cdot \frac{x_2^*}{Q^*} \cdot \frac{\Sigma_S}{\Sigma_S + \Sigma_D}, \quad (89)$$

$$\beta_{C_3} = H^* \cdot \frac{x_3^*}{Q^*} \cdot \frac{\Sigma_S}{\Sigma_S + \Sigma_D}. \quad (90)$$

These five shares sum to H^* by construction.

D. Pigouvian benchmark and contribution in symbols

Let the unregulated equilibrium satisfy

$$S(p^*) = D(p^*), \quad Q^* = S(p^*). \quad (91)$$

Let the welfare optimum solve

$$p_D(Q^{\text{soc}}) = p_S(Q^{\text{soc}}) + MH(Q^{\text{soc}}), \quad (92)$$

with Pigouvian schedule $t(Q) = MH(Q)$ and corresponding prices

$$p_c^{\text{soc}} = p_D(Q^{\text{soc}}), \quad (93)$$

$$p_p^{\text{soc}} = p_S(Q^{\text{soc}}), \quad (94)$$

$$p_c^{\text{soc}} - p_p^{\text{soc}} = MH(Q^{\text{soc}}). \quad (95)$$

Avoidable harm is

$$\Delta H = H(Q^*) - H(Q^{\text{soc}}). \quad (96)$$

Pigouvian causal contribution for each actor a is then

$$B_a = \Delta H \cdot \frac{\Delta \Pi_a}{\sum_{b \in \mathcal{N}} \Delta \Pi_b}, \quad (97)$$

where (using (63) and (65) for surplus)

$$\Delta CS_j = CS_j(p^*) - CS_j(p_c^{\text{soc}}), \quad j \in \{1, 2, 3\}, \quad (98)$$

$$\Delta PS_i = PS_i(p^*) - PS_i(p_p^{\text{soc}}), \quad i \in \{1, 2\}. \quad (99)$$

E. Discussion: realized-harm vs avoidable-harm objects

In the two-producer/three-consumer case, Shapley attribution and elasticity pivotality both allocate $H(Q^*)$ across five players, but they operationalize “pivotality” differently: Shapley averages marginal contributions across coalition orderings, while pivotality allocates by local marginal influence on Q^* under proportional curve perturbations. Pigouvian contribution allocates ΔH (the difference between the unregulated and internalized outcomes). These objects answer different questions, all of which are causal-contribution questions rather than blame judgments (cf. Section I-A):

- **Shapley:** “Who causally contributes to the harm that actually occurs, under an order-averaged coalition notion?”
- **Elasticity pivotality:** “Who is locally pivotal for the realized quantity, as measured by marginal comparative statics at the observed equilibrium?”
- **Pigouvian:** “Who benefits from, and thus causally contributes to, the harm that could be avoided by internalization?”

A complete normative account of blame would need to allocate both $H(Q^{\text{soc}})$ and ΔH under some further, fault-sensitive principle – this paper provides the causal-contribution decomposition on which such a principle could be layered, but does not itself supply that principle.

VIII. CONCLUSION

This paper proposed three curve-based frameworks for distributing causal contribution to harmful market externalities. Shapley attribution provides an axiomatic, coalition-based allocation of realized harm, while elasticity pivotality provides a local (comparative-statics) allocation of realized harm that can be expressed transparently in terms of slopes and elasticities at the observed equilibrium. Pigouvian-counterfactual analysis uses the welfare-optimal benchmark to isolate avoidable harm and allocates causal contribution to it according to private gains from underpricing. The derivations were given for generic (nonlinear) individual supply and demand functions and illustrated symbolically for a small asymmetric market. As discussed in Section I-A, all three measures quantify causal contribution to an outcome; none of them, on its own, constitutes a judgment of moral blame, which would additionally require accounting for fault, foreseeability, and the availability of less harmful alternatives.

Several extensions are immediate: incorporating heterogeneous emissions intensity and avoidable-technology choice, adding abatement choices, treating uncertainty formally via Bayesian equilibria, evaluating alternative welfare criteria (rights-based constraints, caps, distributional objectives) that change the definition of the “acceptable” benchmark harm, and developing a fault-sensitive layer on top of the causal-contribution measures presented here that would be needed to move from contribution to genuine blame.